Glycopyrrolate (glycopyrronium) (systemic): Drug information - UpToDate

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For additional information see "Glycopyrrolate (glycopyrronium) (systemic): Patient drug information" and "Glycopyrrolate (glycopyrronium) (systemic): Pediatric drug information"

For abbreviations, symbols, and age group definitions show table

Brand Names: US

- Cuvposa;
- Dartisla ODT [DSC];
- Glycate;
- Glyrx-PF;
- Robinul [DSC];
- Robinul-Forte [DSC]

Brand Names: Canada

- Cuvposa

Pharmacologic Category

- Anticholinergic Agent

Dosing: Adult

Note: Orally disintegrating tablet 1.7 mg = Regular tablet 2 mg. Dartisla ODT has been discontinued in the United States for >1 year.

Hyperhidrosis, primary focal

Hyperhidrosis, primary focal (off-label use): Limited data available: Oral: 1 to 2 mg once or twice daily may be a typical effective dosage range (Ref); titrate per response and tolerability; doses up to 6 to 8 mg per day in 2 or 3 divided doses have been used in some patients (Ref).

Reduction of secretions

Reduction of secretions (preoperative): IM: 4 mcg/kg 30 to 60 minutes before anesthesia or when the preanesthetic opioid and/or sedative are administered

Reduction of secretions at end of life

Reduction of secretions at end of life (off-label use):

Note: Does not dry secretions already present and typically ineffective for lower respiratory tract secretions (Ref).

IV, SUBQ: 0.1 to 0.2 mg every 4 to 8 hours as needed; do not exceed 4 doses/day (Ref) **or** 0.1 to 0.2 mg once followed by a continuous infusion of 0.6 to 1.2 mg over 24 hours (Ref). Increasing the starting dose to 0.4 mg has been described; however, some experts do not recommend doses above 0.2 mg (Ref).

Sublingual: 0.1 mg (0.5 mL) every 6 hours as needed. **Note:** Administer using the 1 mg per 5 mL oral solution (Ref).

Reversal of bradycardia, vagal reflexes

Reversal of bradycardia, vagal reflexes (intraoperative): IV: 0.1 mg as a single dose; repeat as needed at 2- to 3-minute intervals.

Reversal of muscarinic effects of cholinergic agents

Reversal of muscarinic effects of cholinergic agents: IV: 0.2 mg for each 1 mg of neostigmine or 5 mg of pyridostigmine administered.

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Adult

There are no specific dosage adjustments provided in the manufacturer's labeling; elimination is severely impaired in renal failure; use with caution; dosage adjustment may be necessary.

Dosing: Liver Impairment: Adult

There are no specific dosage adjustments provided in the manufacturer's labeling; use with caution; consider dose reduction.

Dosing: Older Adult

Refer to adult dosing. Use with caution; consider dose reduction.

Dosing: Pediatric

(For additional information see "Glycopyrrolate (glycopyrronium) (systemic): Pediatric drug information")

Dosage guidance:

Safety: Dosing may be presented as mcg/kg, **mg**/**kg**, or **mg**/dose; use extra caution.

Dosage form information: Dartisla ODT has been discontinued in the United States for >1 year.

Chronic drooling

Chronic drooling: Children ≥3 years and Adolescents ≤16 years: Oral solution (Cuvposa): Initial: 20 mcg/kg/dose 3 times daily; titrate in increments of 20 mcg/kg/dose every 5 to 7 days as tolerated to response up to a maximum dose of 100 mcg/kg/dose 3 times daily, not to exceed 1,500 to 3,000 mcg/dose.

Hyperhidrosis, primary focal

Hyperhidrosis, primary focal: Limited data available; optimal dose not established:

Children ≥8 years and Adolescents: Oral: Usual dose: 1 to 2 **mg** once or twice daily; titrate per response and tolerability; reported range: 0.5 to 6 **mg** per **day** in divided doses one to three times daily (Ref).

Rapid sequence intubation, pretreatment

Rapid sequence intubation, pretreatment:

Infants, Children, and Adolescents: IV, Intraosseous: 0.005 to 0.01 **mg/kg**; maximum dose: 0.2 **mg**/dose (Ref).

Reduction of respiratory secretions, palliative care

Reduction of respiratory secretions, palliative care: Limited data available: Children: Oral: 40 to 100 mcg/kg/dose every 4 to 8 hours (Ref).

Reduction of secretions, preoperative

Reduction of secretions, preoperative:

Infants and Children ≤2 years: IM: 4 to 9 mcg/kg as a single dose 30 to 60 minutes before procedure.

Children >2 years and Adolescents: IM: 4 mcg/kg as a single dose 30 to 60 minutes before procedure.

Reversal of bradycardia, vagal reflexes

Reversal of bradycardia, vagal reflexes (intraoperative): Infants, Children, and Adolescents: IV: 4 mcg/kg/dose (maximum dose: 100 mcg/dose) at 2- to 3-minute intervals.

Reversal of muscarinic effects of cholinergic agents

Reversal of muscarinic effects of cholinergic agents: Infants, Children, and Adolescents: IV: 0.2 **mg** for each 1 mg of neostigmine or 5 mg of pyridostigmine administered.

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Pediatric

There are no specific dosage adjustments provided in the manufacturer's labeling; elimination is severely impaired in renal failure; use with caution; dosage adjustment may be necessary.

Dosing: Liver Impairment: Pediatric

There are no specific dosage adjustments provided in the manufacturer's labeling (has not been studied); use with caution.

Adverse Reactions (Significant): Considerations

Anticholinergic effects

Anticholinergic agents, including systemic glycopyrrolate, may cause reversible, dose-dependent anticholinergic effects, including, but not limited to, CNS effects (eg, **behavioral changes** [eg, **irritability**, **agitation**, **dizziness, headache**]); genitourinary effects (eg, **urinary retention**); GI effects (eg, **xerostomia, constipation, gastrointestinal pseudo-obstruction**); ophthalmic effects (eg, **blurred vision, mydriasis**); cardiovascular effects (eg, **palpitations**); and dermatologic effects (eg, hypohidrosis) in adult and pediatric patients (Ref). **Xerostomia** is the most frequent adverse reaction resulting in discontinuation (Ref).

Mechanism: Dose-related; an extension of antimuscarinic pharmacological properties (Ref).

Risk factors:

• Higher doses (Ref)

• Concurrent use of other anticholinergic medications

• Older age

GI effects

Various GI effects may occur with systemic glycopyrrolate, including **xerostomia, constipation,** and **gastrointestinal pseudo-obstruction** in adult and pediatric patients (Ref). GI pseudo-obstruction may result in abdominal distention, pain, nausea, or **vomiting**. **Diarrhea** may be an early sign of incomplete GI obstruction, especially in patients with an ileostomy or colostomy.

Mechanism: Dose-related; an extension of antimuscarinic pharmacological properties (Ref).

Risk factors:

• Mechanical GI obstructive diseases (contraindicated)

• GI motility disorders (contraindicated)

• Concurrent use of medications that affect GI peristalsis

Heat prostration

Anticholinergic agents, including systemic glycopyrrolate, may cause hypohidrosis, which may result in life-threatening hyperthermia or heatstroke (Ref).

Mechanism: Dose-related; related to the pharmacologic action. Antagonism of central and peripheral muscarinic receptors results in increased heat production due to increased muscle activity and impaired sweat gland function (Ref).

Risk factors:

• Presence of fever

• Physical exertion in high ambient temperature and humidity (Ref)

• Concurrent use of other anticholinergic medications (Ref)

• Children and older adults

Adverse Reactions

The following adverse drug reactions and incidences are derived from product labeling unless otherwise specified. Incidences reported with oral use of glycopyrrolate in children and adolescents.

>10%:

Cardiovascular: Flushing (30%)

Gastrointestinal: Constipation (35%) (table 1)Constipation, vomiting (40%) (table 2)Vomiting, xerostomia (40%) (table 3)Xerostomia

Drug (Glycopyrrolate)	Placebo	Population	Dose
35%	22%	Pediatric patients 3 to 16 years	20 mcg/kg/dose 3 times daily, titrate in

Drug (Glycopyrrolate)	Placebo	Population	Dose
40%	11%	Pediatric patients 3 to 16 years	20 mcg/kg/dose 3 times daily, titrate in

Drug (Glycopyrrolate)	Placebo	Population	Dose
40%	11%	Pediatric patients 3 to 16 years	20 mcg/kg/dose 3 times daily, titrate in

Genitourinary: Urinary retention (15%) (table 4)Urinary Retention

Drug (Glycopyrrolate)	Placebo	Population	Dose
15%	0%	Pediatric patients 3 to 16 years	20 mcg/kg/dose 3 times daily, titrate in

Nervous system: Headache (15%) (table 5)Headache

Drug (Glycopyrrolate)	Placebo	Population	Dose
15%	6%	Pediatric patients 3 to 16 years	20 mcg/kg/dose 3 times daily, titrate in

Respiratory: Nasal congestion (30%), sinusitis (15%), upper respiratory tract infection (15%)

Postmarketing (all routes of administration and ages):

Cardiovascular: Cardiac arrhythmia (including bradycardia, ventricular fibrillation, ventricular tachycardia), hypertension, hypotension, palpitations (Ref)

Dermatologic: Cheilosis (Ref)

Gastrointestinal: Ageusia, diarrhea (Ref), gastrointestinal pseudo-obstruction (Ref), halitosis (Ref)

Genitourinary: Lactation insufficiency (suppression)

Hypersensitivity: Facial swelling (Ref)

Local: Injection-site reaction (including erythema at injection site, injection-site pruritus, localized edema, pain at injection site)

Nervous system: Agitation (Ref), behavioral changes (Ref), dizziness (Ref), irritability (Ref), personality changes (Ref), seizure

Ophthalmic: Blurred vision (Ref), mydriasis (Ref)

Respiratory: Epistaxis (Ref)

Miscellaneous: Fever (Ref), malignant hyperthermia

Contraindications

Hypersensitivity to glycopyrrolate or any component of the formulation; medical conditions that preclude use of anticholinergic medication (eg, severe ulcerative colitis, toxic megacolon complicating ulcerative colitis, paralytic ileus, obstructive disease of GI tract [eg, achalasia, pyloroduodenal stenosis, strictures], bleeding GI ulcer, intestinal atony in older adult or debilitated patients, unstable cardiovascular status in acute hemorrhage, glaucoma, obstructive uropathy [eg, bladder neck obstruction due to prostatic hypertrophy], myasthenia gravis)

Oral solution: Additional contraindication: Concomitant use of potassium chloride in a solid oral dosage form.

Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Canadian labeling: Additional contraindications (not in US labeling): Injection: Use in newborns; chronic lung disease in older adult or debilitated patients.

Warnings/Precautions

Disease-related concerns:

• Cardiovascular disease: Use with caution in patients with coronary artery disease, arrhythmias, heart failure, hypertension, or tachycardia; evaluate tachycardia prior to administration.

• Hepatic impairment: Use with caution in patients with hepatic impairment; consider dosage reduction.

• Hiatal hernia: Use with caution in patients with hiatal hernia with reflux esophagitis; consider dosage reduction.

• Hyperthyroidism: Use with caution in patients with hyperthyroidism.

• Neuropathy: Use with caution in patients with autonomic neuropathy; consider dosage reduction.

• Prostatic hyperplasia/bladder neck obstruction: May worsen the symptoms (eg, urinary retention) of prostatic hyperplasia and/or bladder neck obstruction; use with caution; consider dosage reduction in patients with prostatic hypertrophy.

• Renal impairment: Use with caution in patients with renal impairment; elimination is severely impaired in renal failure; dosage adjustment may be necessary.

• Ulcerative colitis: Use with caution in patients with ulcerative colitis; large doses may suppress intestinal motility; consider dosage reduction; may precipitate/exacerbate an ileus or toxic megacolon. Use is contraindicated in patients with severe ulcerative colitis.

Special populations:

• Older adult: Use with caution in older adults; increased risk for anticholinergic effects, confusion, and hallucinations; consider dosage reduction.

• Pediatric: Infants, patients with Down syndrome, and pediatric patients with spastic paralysis or brain damage may experience an increased response to anticholinergics, increasing the potential for adverse events. A paradoxical reaction characterized by hyperexcitability may occur in pediatric patients taking large doses. Infants and young children are especially susceptible to the toxic effects of anticholinergics.

Dosage form specific issues:

• Benzyl alcohol and derivatives: Some dosage forms may contain benzyl alcohol; large amounts of benzyl alcohol (≥99 mg/kg/day) have been associated with a potentially fatal toxicity (“gasping syndrome”) in neonates; the “gasping syndrome” consists of metabolic acidosis, respiratory distress, gasping respirations, CNS dysfunction (including convulsions, intracranial hemorrhage), hypotension and cardiovascular collapse (AAP [”Inactive” 1997]; CDC 1982); some data suggests that benzoate displaces bilirubin from protein binding sites (Ahlfors 2001); avoid or use dosage forms containing benzyl alcohol with caution in neonates. See manufacturer’s labeling.

• Injection: Sensitivity of the eyes to light may occur. Protect eyes from light after administration.

• Propylene glycol: Some dosage forms may contain propylene glycol; large amounts are potentially toxic and have been associated hyperosmolality, lactic acidosis, seizures, and respiratory depression; use caution (AAP 1997; Zar 2007).

Warnings: Additional Pediatric Considerations

Some dosage forms may contain propylene glycol; in neonates large amounts of propylene glycol delivered orally, intravenously (eg, >3,000 mg/day), or topically have been associated with potentially fatal toxicities which can include metabolic acidosis, seizures, renal failure, and CNS depression; toxicities have also been reported in children and adults including hyperosmolality, lactic acidosis, seizures, and respiratory depression; use caution (AAP 1997; Shehab 2009).

Product Availability

Dartisla ODT has been discontinued in the United States for >1 year.

Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling. [DSC] = Discontinued product

Solution, Injection:

Glyrx-PF: 0.2 mg/mL (1 mL); 0.4 mg/2 mL (2 mL)

Generic: 0.2 mg/mL (1 mL); 0.4 mg/2 mL (2 mL); 1 mg/5 mL (5 mL); 4 mg/20 mL (20 mL)

Solution, Injection [preservative free]:

Generic: 0.2 mg/mL (1 mL); 0.4 mg/2 mL (2 mL)

Solution, Oral:

Cuvposa: 1 mg/5 mL (473 mL) [contains methylparaben (methyl p-hydroxybenzoate), propylene glycol, propylparaben (propyl p-hydroxybenzoate), saccharin sodium; cherry flavor]

Generic: 1 mg/5 mL (473 mL)

Solution Prefilled Syringe, Injection:

Generic: 0.6 mg/3 mL (3 mL)

Solution Prefilled Syringe, Injection [preservative free]:

Glyrx-PF: 0.6 mg/3 mL (3 mL [DSC]); 1 mg/5 mL (5 mL)

Generic: 0.2 mg/mL (1 mL); 0.4 mg/2 mL (2 mL); 0.6 mg/3 mL (3 mL)

Tablet, Oral:

Glycate: 1.5 mg [dye free]

Robinul: 1 mg [DSC] [scored; dye free]

Robinul-Forte: 2 mg [DSC] [scored; dye free]

Generic: 1 mg, 1.5 mg, 2 mg

Tablet Disintegrating, Oral:

Dartisla ODT: 1.7 mg [DSC] [contains gelatin (fish)]

Generic Equivalent Available: US

May be product dependent

Pricing: US

Solution (Cuvposa Oral)

1 mg/5 mL (per mL): $1.27

Solution (Glycopyrrolate Injection)

0.2 mg/mL (per mL): $0.85 - $16.25

0.4 mg/2 mL (per mL): $0.75 - $14.38

1 mg/5 mL (per mL): $0.66 - $11.00

4 mg/20 mL (per mL): $1.44 - $8.93

Solution (Glycopyrrolate Oral)

1 mg/5 mL (per mL): $0.43 - $1.27

Solution (Glyrx-PF Injection)

0.2 mg/mL (per mL): $4.56

0.4 mg/2 mL (per mL): $3.96

Solution Prefilled Syringe (Glycopyrrolate PF Injection)

0.2 mg/mL (per mL): $7.20 - $8.38

0.4 mg/2 mL (per mL): $6.60 - $8.33

0.6 mg/3 mL (per mL): $6.11

Solution Prefilled Syringe (Glyrx-PF Injection)

1 mg/5 mL (per mL): $3.84

Tablets (Glycate Oral)

1.5 mg (per each): $19.90

Tablets (Glycopyrrolate Oral)

1 mg (per each): $1.06 - $1.31

1.5 mg (per each): $41.39

2 mg (per each): $1.76 - $2.20

Disclaimer: A representative AWP (Average Wholesale Price) price or price range is provided as reference price only. A range is provided when more than one manufacturer's AWP price is available and uses the low and high price reported by the manufacturers to determine the range. The pricing data should be used for benchmarking purposes only, and as such should not be used alone to set or adjudicate any prices for reimbursement or purchasing functions or considered to be an exact price for a single product and/or manufacturer. Medi-Span expressly disclaims all warranties of any kind or nature, whether express or implied, and assumes no liability with respect to accuracy of price or price range data published in its solutions. In no event shall Medi-Span be liable for special, indirect, incidental, or consequential damages arising from use of price or price range data. Pricing data is updated monthly.

Dosage Forms: Canada

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Solution, Injection:

Generic: 0.2 mg/mL (1 mL, 2 mL, 20 mL); 0.4 mg/2 mL (2 mL); 4 mg/20 mL (20 mL)

Solution, Oral:

Cuvposa: 1 mg/5 mL (473 mL) [contains methylparaben (methyl p-hydroxybenzoate), propylene glycol, propylparaben (propyl p-hydroxybenzoate), saccharin sodium, sorbitol]

Extemporaneous Preparations

Note: A glycopyrrolate oral solution (0.2 mg/mL) is commercially available.

0.5 mg/mL Oral Suspension

A 0.5 mg/mL oral suspension may be made with 1 mg tablets and a 1:1 mixture of Ora-Plus and either Ora-Sweet or Ora-Sweet SF. Crush thirty 1 mg tablets in a mortar and reduce to a fine powder. Prepare diluent by mixing 30 mL of Ora-Plus with 30 mL of either Ora-Sweet or Ora-Sweet SF and stir vigorously. Add 30 mL of diluent (via geometric dilution) to powder until smooth suspension is obtained. Transfer suspension to 60 mL amber bottle. Rinse contents of mortar into bottle with sufficient quantity of remaining diluent to obtain 60 mL (final volume). Label "shake well". Stable at room temperature for 90 days. Due to bitter aftertaste, chocolate syrup may be administered prior to or mixed (1:1 v/v) with suspension immediately before administration (Ref).

A 0.5 mg/mL oral solution can be made from tablets. Crush fifty 1 mg tablets in a mortar and reduce to a fine powder. Add enough distilled water to make about 90 mL, mix well. Transfer to a bottle, rinse mortar with water, and add a quantity of water sufficient to make 100 mL. Label "shake well" and "protect from light". Stable at room temperature for 25 days (Ref).

0.1 mg/mL Oral Solution

A 0.1 mg/mL oral solution may be made using glycopyrrolate 0.2 mg/mL injection without preservatives. Withdraw 50 mL from vials with a needle and syringe, add to 50 mL of a 1:1 mixture of Ora-Sweet and Ora-Plus in a bottle. Label “shake well”, “protect from light,” and “refrigerate”. Stable refrigerated for 35 days (Ref).

Administration: Adult

IM: May administer undiluted.

IV: May be administered IV without dilution or may dilute in a compatible solution. In perioperative setting, usually administered over 1 to 2 minutes (eg, in adults: 0.2 mg over 1 to 2 minutes). May also be administered via the tubing of a running IV infusion of a NS. May be administered IV in the same syringe with neostigmine or pyridostigmine for reversal of neuromuscular blockade.

Oral:

Orally disintegrating tablet: Administer at least 1 hour before or 2 hours after food. Remove tablet from blister packaging immediately prior to administration; do not push tablet through foil. Place tablet on tongue and allow to disintegrate then swallow without water. **Tablets should not be broken or cut.**

Oral solution: Administer 1 hour before or 2 hours after meals. Measure oral solution with an accurate measuring device (eg, dosing cup, oral syringe).

SUBQ (off-label route): May administer SUBQ (Ref).

Preparation for Administration: Adult

IV: May further dilute in D5W, D10W, NS, D5NS, or D5$^{1}/_{2}$NS.

Administration: Pediatric

Oral:

Tablets: Administer without regard to meals.

Oral solution: Administer on an empty stomach, 1 hour before or 2 hours after meals; measure with an accurate measuring device (eg, dosing cup, oral syringe).

Parenteral: May be administered IM (undiluted) or by slow IV injection (with or without dilution in a compatible fluid); in perioperative setting, usually administered over 1 to 2 minutes (eg, in adults: 0.2 mg over 1 to 2 minutes). May also be administered via the tubing of a running IV infusion of NS. May be administered IV in the same syringe with neostigmine or pyridostigmine for reversal of neuromuscular blockade.

Preparation for Administration: Pediatric

Parenteral: Slow IV injection: May further dilute in D5W, D10W, NS, D5NS, D5$^{1}/_{2}$NS.

Storage/Stability

Injection: Store at 20°C to 25°C (68°F to 77°F).

Oral: Store at 20°C to 25°C (68°F to 77°F); excursions permitted to 15°C to 30°C (59°F to 86°F).

Use: Labeled Indications

Chronic drooling (Cuvposa only): To reduce chronic, severe drooling in pediatric patients 3 to 16 years with neurologic conditions (eg, cerebral palsy) associated with problem drooling

Reduction of secretions (injection only): To reduce salivary, tracheobronchial, and pharyngeal secretions and to reduce the volume and acidity of gastric secretions during induction of anesthesia and intubation

Reversal of bradycardia, vagal reflexes (injection only): To block cardiac vagal inhibitory reflexes during induction of anesthesia and intubation; intraoperatively to counteract surgically or drug-induced or vagal reflexes associated arrhythmias

Reversal of muscarinic effects of cholinergic agents (injection only): Protects against the peripheral muscarinic effects (eg, bradycardia, excessive secretions) of cholinergic agents (eg, neostigmine, pyridostigmine) given to reverse the neuromuscular blockade due to non-depolarizing muscle relaxants

Use: Off-Label: Adult

Reduction of secretions at end of life; Primary focal hyperhidrosis

Medication Safety Issues

High alert medication

The Institute for Safe Medication Practices (ISMP) includes this medication among its list of drugs (pediatric liquid medications requiring measurement) which have a heightened risk of causing significant patient harm when used in error (ISMP List of High-Alert Medications in Community/Ambulatory Care Settings ©ISMP 2021).

International issues:

Robinul [US and multiple international markets] may be confused with Reminyl brand name for galantamine [Canada and multiple international markets]

Metabolism/Transport Effects

Substrate of MATE1/2-K

Drug Interactions

(For additional information: Launch drug interactions program)

Note: Interacting drugs may **not be individually listed below** if they are part of a group interaction (eg, individual drugs within "CYP3A4 Inducers [Strong]" are NOT listed). For a complete list of drug interactions by individual drug name and detailed management recommendations, use the drug interactions program by clicking on the "Launch drug interactions program" link above.

Acetylcholinesterase Inhibitors: May decrease therapeutic effects of Agents with Clinically Relevant Anticholinergic Effects. Agents with Clinically Relevant Anticholinergic Effects may decrease therapeutic effects of Acetylcholinesterase Inhibitors. *Risk C: Monitor*

Aclidinium: May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk X: Avoid*

Acrivastine: May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk C: Monitor*

Agents with Clinically Relevant Anticholinergic Effects: May increase anticholinergic effects of Glycopyrrolate (Systemic). *Risk C: Monitor*

Amantadine: May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk C: Monitor*

Atenolol: Glycopyrrolate (Systemic) may increase serum concentrations of Atenolol. *Risk C: Monitor*

Benperidol: Agents with Clinically Relevant Anticholinergic Effects may decrease therapeutic effects of Benperidol. *Risk C: Monitor*

Benztropine: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Benztropine. *Risk C: Monitor*

Biperiden: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Biperiden. *Risk C: Monitor*

Bornaprine: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Bornaprine. *Risk C: Monitor*

Botulinum Toxin-Containing Products: May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk C: Monitor*

Bromperidol: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Bromperidol. *Risk C: Monitor*

Buclizine: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Buclizine. *Risk C: Monitor*

Cannabinoid-Containing Products: Agents with Clinically Relevant Anticholinergic Effects may increase tachycardic effects of Cannabinoid-Containing Products. *Risk C: Monitor*

Chlorprothixene: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Chlorprothixene. *Risk C: Monitor*

Cimetropium: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Cimetropium. *Risk X: Avoid*

CloZAPine: Agents with Clinically Relevant Anticholinergic Effects may increase constipating effects of CloZAPine. Management: Consider alternatives to this combination whenever possible. If combined, monitor closely for signs and symptoms of gastrointestinal hypomotility and consider prophylactic laxative treatment. *Risk D: Consider Therapy Modification*

Cyclizine: May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk C: Monitor*

Darifenacin: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Darifenacin. *Risk C: Monitor*

Dicyclomine: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Dicyclomine. *Risk C: Monitor*

Digoxin: Glycopyrrolate (Systemic) may increase serum concentrations of Digoxin. *Risk C: Monitor*

Dimethindene (Systemic): Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Dimethindene (Systemic). *Risk C: Monitor*

DroNABinol: Agents with Clinically Relevant Anticholinergic Effects may increase tachycardic effects of DroNABinol. *Risk X: Avoid*

Eluxadoline: Agents with Clinically Relevant Anticholinergic Effects may increase constipating effects of Eluxadoline. *Risk X: Avoid*

Fesoterodine: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Fesoterodine. *Risk C: Monitor*

FluPHENAZine: May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk C: Monitor*

Gastrointestinal Agents (Prokinetic): Agents with Clinically Relevant Anticholinergic Effects may decrease therapeutic effects of Gastrointestinal Agents (Prokinetic). *Risk C: Monitor*

Gepotidacin: May decrease anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk C: Monitor*

Glucagon: Agents with Clinically Relevant Anticholinergic Effects may increase adverse/toxic effects of Glucagon. Specifically, the risk of gastrointestinal adverse effects may be increased. *Risk C: Monitor*

Glycopyrrolate (Oral Inhalation): Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Glycopyrrolate (Oral Inhalation). *Risk X: Avoid*

Glycopyrronium (Topical): May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk X: Avoid*

Haloperidol: Glycopyrrolate (Systemic) may decrease serum concentrations of Haloperidol. Management: Consider avoiding concurrent use of glycopyrrolate and haloperidol.Monitor patients closely for signs/symptoms of reduced clinical response to haloperidol if concurrent use with glycopyrrolate is required. *Risk D: Consider Therapy Modification*

Hyoscyamine: Anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects and Hyoscyamine may increase with coadministration. *Risk C: Monitor*

Ipratropium (Nasal): May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk C: Monitor*

Ipratropium (Oral Inhalation): May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk X: Avoid*

Itopride: Agents with Clinically Relevant Anticholinergic Effects may decrease therapeutic effects of Itopride. *Risk C: Monitor*

Levodopa-Foslevodopa: Glycopyrrolate (Systemic) may decrease serum concentrations of Levodopa-Foslevodopa. *Risk C: Monitor*

Levosulpiride: Agents with Clinically Relevant Anticholinergic Effects may decrease therapeutic effects of Levosulpiride. *Risk X: Avoid*

Melperone: May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk C: Monitor*

MetFORMIN: Glycopyrrolate (Systemic) may increase serum concentrations of MetFORMIN. *Risk C: Monitor*

Methotrimeprazine: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Methotrimeprazine. *Risk C: Monitor*

Methscopolamine: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Methscopolamine. *Risk C: Monitor*

Mirabegron: Agents with Clinically Relevant Anticholinergic Effects may increase adverse/toxic effects of Mirabegron. *Risk C: Monitor*

Nitroglycerin: Agents with Clinically Relevant Anticholinergic Effects may decrease absorption of Nitroglycerin. Specifically, anticholinergic agents may decrease the dissolution of sublingual nitroglycerin tablets, possibly impairing or slowing nitroglycerin absorption. *Risk C: Monitor*

OLANZapine: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of OLANZapine. *Risk C: Monitor*

Opioid Agonists: Agents with Clinically Relevant Anticholinergic Effects may increase adverse/toxic effects of Opioid Agonists. Specifically, the risk for constipation and urinary retention may be increased with this combination. *Risk C: Monitor*

Opipramol: May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk C: Monitor*

Oxatomide: May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk X: Avoid*

OxyBUTYnin: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of OxyBUTYnin. *Risk C: Monitor*

Perazine: May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk C: Monitor*

Periciazine: May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk C: Monitor*

Perphenazine: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Perphenazine. *Risk C: Monitor*

Potassium Chloride: Agents with Clinically Relevant Anticholinergic Effects may increase ulcerogenic effects of Potassium Chloride. Management: Patients on drugs with substantial anticholinergic effects should avoid using any solid oral dosage form of potassium chloride. *Risk X: Avoid*

Potassium Citrate: Agents with Clinically Relevant Anticholinergic Effects may increase ulcerogenic effects of Potassium Citrate. Management: Patients on drugs with substantial anticholinergic effects should avoid using any solid oral dosage form of potassium citrate. *Risk X: Avoid*

Pramlintide: May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. These effects are specific to the GI tract. *Risk X: Avoid*

Promethazine: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Promethazine. *Risk C: Monitor*

Propantheline: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Propantheline. *Risk C: Monitor*

Propiverine: May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk C: Monitor*

QuiNIDine: May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk C: Monitor*

Ramosetron: Agents with Clinically Relevant Anticholinergic Effects may increase constipating effects of Ramosetron. *Risk C: Monitor*

Revefenacin: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Revefenacin. *Risk X: Avoid*

Rivastigmine: Agents with Clinically Relevant Anticholinergic Effects may decrease therapeutic effects of Rivastigmine. Rivastigmine may decrease therapeutic effects of Agents with Clinically Relevant Anticholinergic Effects. Management: Use of rivastigmine with an anticholinergic agent is not recommended unless clinically necessary. If the combination is necessary, monitor for reduced anticholinergic effects. *Risk D: Consider Therapy Modification*

Scopolamine: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Scopolamine. *Risk C: Monitor*

Secretin: Agents with Clinically Relevant Anticholinergic Effects may decrease therapeutic effects of Secretin. Management: Avoid concomitant use of anticholinergic agents and secretin. Discontinue anticholinergic agents at least 5 half-lives prior to administration of secretin. *Risk D: Consider Therapy Modification*

Sofpironium: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Sofpironium. *Risk X: Avoid*

Thiazide and Thiazide-Like Diuretics: Agents with Clinically Relevant Anticholinergic Effects may increase serum concentrations of Thiazide and Thiazide-Like Diuretics. *Risk C: Monitor*

Thiothixene: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Thiothixene. *Risk C: Monitor*

Tiapride: Agents with Clinically Relevant Anticholinergic Effects may decrease therapeutic effects of Tiapride. *Risk C: Monitor*

Tiotropium: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Tiotropium. *Risk X: Avoid*

Tolterodine: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Tolterodine. *Risk C: Monitor*

Topiramate: Agents with Clinically Relevant Anticholinergic Effects may increase adverse/toxic effects of Topiramate. *Risk C: Monitor*

Trimethobenzamide: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Trimethobenzamide. *Risk C: Monitor*

Trospium: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Trospium. *Risk C: Monitor*

Umeclidinium: May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk X: Avoid*

Zuclopenthixol: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Zuclopenthixol. *Risk C: Monitor*

Food Interactions

Administration with a high-fat meal significantly reduced absorption. Management: Administer at least 1 hour before or 2 hours after meals.

Pregnancy Considerations

When given as a single maternal injection prior to delivery, glycopyrrolate was not found to cross the placenta in significant amounts. In addition, fetal heart rate, fetal heart rate variability, and Apgar scores were not significantly affected (Abboud 1981; Abboud 1983; Ali-Melkkilä 1990; Ure 1999).

Use of topical glycopyrrolate for the treatment of hyperhidrosis in a pregnant patient diagnosed with herpetic neuralgia has been described in a case report (Ladjevic 2009).

Breastfeeding Considerations

It is not known if glycopyrrolate is present in breast milk.

According to the manufacturer, the decision to breastfeed during therapy should consider the risk of infant exposure, the benefits of breastfeeding to the infant, and benefits of treatment to the mother. Glycopyrrolate is an anticholinergic and may suppress lactation.

Monitoring Parameters

Heart rate; anticholinergic effects; bowel sounds; bowel movements; effects on drooling

Mechanism of Action

Blocks the action of acetylcholine at parasympathetic sites in smooth muscle, secretory glands, and the CNS; indirectly reduces the rate of salivation by preventing the stimulation of acetylcholine receptors

Pharmacokinetics (Adult Data Unless Noted)

Onset of action: IM: 15 to 30 minutes; IV: Within 1 minute.

Peak effect: IM: Within ~30 to 45 minutes.

Duration: Vagal effect: 2 to 3 hours; Inhibition of salivation: Up to 7 hours; Parenteral: 7 hours.

Absorption: Oral tablet: Poor (~3%); variable and erratic; Oral solution: 23% lower compared to tablet; high-fat meal markedly reduces the oral bioavailability.

Distribution: V_d: IV: Children and Adolescents ≤14 years: 1.3 to 1.8 L/kg (range: 0.7 to 3.9 L/kg); Adults: 0.42 ± 0.22 L/kg.

Bioavailability: Tablet: 3%.

Half-life elimination: IV: Infants: 21.6 to 130 minutes; Children: 19.2 to 99.2 minutes; IM: Adults: 0.55 to 1.25 hours; IV: 0.83 ± 0.27 hour; Oral solution: Adults: 3 hours; Orally disintegrating tablet: Adults: 2.8 hours.

Time to peak, plasma: IM: ~30 minutes; Oral solution: 3.1 hours; Orally disintegrating tablet: Median: 3 hours.

Excretion: Urine (as unchanged drug, IM: >80%, IV: 85%); bile (as unchanged drug, <5%).

Clearance: Children and Adolescents ≤14 years: Mean range: 1.01 to 1.41 L/kg/hour (range: 0.32 to 2.22 L/kg/hour); Adults: 0.54 ± 0.14 L/kg/hour.

Pharmacokinetics: Additional Considerations (Adult Data Unless Noted)

Altered kidney function: Elimination is severely impaired in patients with renal failure.

Patient Counseling Points

(For additional information see ”Glycopyrrolate (glycopyrronium) (systemic): Patient drug information”)

What is this drug used for?

• It is used to help with symptoms of GI (gastrointestinal) ulcers.

• It is used to reduce drooling.

• In surgery, it is used to lower secretions such as saliva.

• It may be given to you for other reasons. Talk with the doctor.

All drugs may cause side effects. However, many people have no side effects or only have minor side effects. Call your doctor or get medical help if any of these side effects or any other side effects bother you or do not go away:

• Dry mouth

• Upset stomach or throwing up

• Stuffy nose

• Feeling dizzy, sleepy, tired, or weak

• Blurred eyesight

• Flushing

• Change in taste

• Headache

• Feeling nervous and excitable

• Trouble sleeping

WARNING/CAUTION: Even though it may be rare, some people may have very bad and sometimes deadly side effects when taking a drug. Tell your doctor or get medical help right away if you have any of the following signs or symptoms that may be related to a very bad side effect:

• Not sweating during activities or in warm temperatures

• Trouble passing urine

• Chest pain or pressure, a fast heartbeat, or an abnormal heartbeat

• Fast breathing

• Fever

• Larger pupils

• Change in eyesight, eye pain, or very bad eye irritation

• Feeling confused

• Diarrhea or constipation

• Bloating

• Swelling of belly

• Stomach pain

• Not able to get or keep an erection

• Muscle weakness

• Signs of an allergic reaction, like rash; hives; itching; red, swollen, blistered, or peeling skin with or without fever; wheezing; tightness in the chest or throat; trouble breathing, swallowing, or talking; unusual hoarseness; or swelling of the mouth, face, lips, tongue, or throat.

Note: This is not a comprehensive list of all side effects. Talk to your doctor if you have questions.

Consumer Information Use and Disclaimer: This information should not be used to decide whether or not to take this medicine or any other medicine. Only the healthcare provider has the knowledge and training to decide which medicines are right for a specific patient. This information does not endorse any medicine as safe, effective, or approved for treating any patient or health condition. This is only a limited summary of general information about the medicine's uses from the patient education leaflet and is not intended to be comprehensive. This limited summary does NOT include all information available about the possible uses, directions, warnings, precautions, interactions, adverse effects, or risks that may apply to this medicine. This information is not intended to provide medical advice, diagnosis or treatment and does not replace information you receive from the healthcare provider. For a more detailed summary of information about the risks and benefits of using this medicine, please speak with your healthcare provider and review the entire patient education leaflet.

Brand Names: International

International Brand Names by Country

For country code abbreviations (show table)

- (AR) Argentina: Robanul;
- (AU) Australia: Robinul;
- (CN) China: Glycopyrrolate;
- (EE) Estonia: Robinul;
- (FI) Finland: Gastrodyn | Robinul | Tarodyl;
- (GB) United Kingdom: Glycopyrronium | Robinul;
- (HK) Hong Kong: Robinul;
- (IE) Ireland: Glycopyrrolate | Robinul;
- (IN) India: Gcolate;
- (JP) Japan: Contrem | Geschnal | Robinul | Sroton;
- (KR) Korea, Republic of: Glycopyrrolate;
- (MY) Malaysia: Glycopyrrolate;
- (NO) Norway: Robinul;
- (NZ) New Zealand: Robinul;
- (PR) Puerto Rico: Dartisla odt | Glycopyrrolate | Glyrx pf | Robinul;
- (SA) Saudi Arabia: Robinul;
- (SE) Sweden: Robinul;
- (TW) Taiwan: Gastrodyn | Robin | Robinul;
- (ZA) South Africa: Robinul

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